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****Abstract****

This report presents a universal and domain-independent medical artificial intelligence (AI) platform designed to enhance surgical decision-making, improve intraoperative patient safety, and optimize healthcare data management across all medical domains. Unlike existing AI systems that are limited to specific procedures or specialties, the proposed platform is built on a fundamental intelligence core that is augmented by modular domain adaptation layers.

The system acts as an intelligent co-pilot for clinical teams, providing real-time risk detection for human and non-human agents, reducing cognitive workload, and supporting coordinated decision-making in connected operating rooms. In addition, the platform introduces advanced mechanisms for data management, privacy, and scalable integration across various surgical and medical environments.

This document represents a research-based proof-of-concept disclosure intended for open scientific publication. Patent and commercialization paths are currently underway.

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****Keywords****

Medical AI;

Surgical decision support;

Patient safety;

Domain-independent AI;

Surgical data science;

Operating room intelligence;

Risk prevention systems;

Healthcare innovation

*Surgical decision support

*Data management and monitoring

*Human and non-human risk prevention

*Surgical and operational efficiency